Churn Analysis
And
Customer Intelligence

1. The Business Challenge

In the hyper-competitive OTT landscape (Netflix, Hotstar, Prime), retention is the only way to survive. You will act as a Data Analyst tasked with identifying high-risk subscribers using a multi-dimensional dataset (Customer demographics, Subscription tiers, and Support escalations).

2. Core Tech Stack

SQL & Python Integration (*numpy*, *pandas*, *sqlite3*, *matplotlib*, *seaborn*), Performance Engineering (*data cleaning*, *feature engineering*, *analytics*), Behavioral Visualization, Writing Actionable Insights

3. Project Milestones

Relational Data Extraction: Connecting Python to SQL databases to pull multi-table datasets.

Advanced Feature Engineering: Data imports, calculating tenure, churn rates, and customer aging.

Executive Reporting: Translating technical findings into billion-dollar business insights.

Connect SQL database to Python – *pandas and sqlite3*

Data import using SQL query in Python – *pandas and sqlite3*

Data Cleaning – *numpy and pandas* data types, rename cols, select specific cols, QCs, handle missing/null values

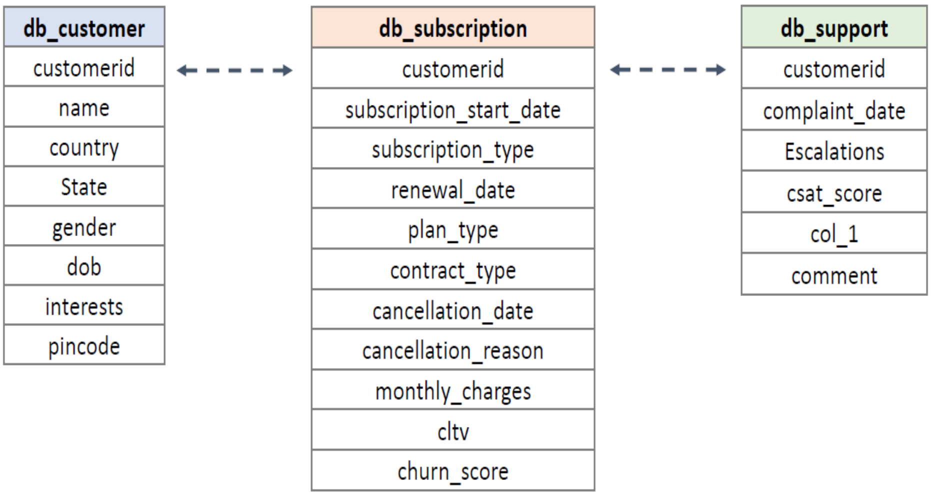
Feature Engineering – *numpy and pandas* Create new calculated cols, data transformation, use filters

Data Analysis – *numpy and pandas* EDA - aggregation, group by, pivot table

Data Visualization – *matplotlib and seaborn*

Database Tables

Database: customer_churn



KPI	Formula
Churn Rate	Churned Customers / Total Customers
Churn by Plan Type	Churn Rate grouped by `plan_type` (Basic, Standard, Premium)
Churn by State	Churn Rate grouped by `country`, `state`
Retention Rate	1 − Churn Rate
ARPU (Average Revenue Per User)	`SUM(monthly_charges) / COUNT(active_customerid)`
Average Customer Tenure	`AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))`
Revenue at Risk	`SUM(monthly_charges) WHERE churn_score > 70`
Escalation Rate	`(SUM(escalations) / COUNT(complaints)) × 100`
Average Complaints per Customer	`COUNT(complaints) / COUNT(DISTINCT customerid)`
Correlation: Escalations → Churn	Compare Churn Rate where `escalations ≥ 1` vs `escalations = 0`

Insights

Insights

Churn Rate: 28.6% | Retention Rate: 71.4%

Most of the churn is from basic subscription plan – nothing to worry in terms of major revenue impact

Most of the churn happened in the month of Sep 2024 and, most affected state is Karnataka

Average Tenure (Days): 1,451 | ARPU is Rs 18.8

Total Revenue = 395

Revenue loss due to churn = 74 | CLTV Lost = 2,047

% Revenue loss = 18%

monthly vs annual churn = 55.6% vs 8.3%

Action Items

Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc

Did we increased subs price for **Basic plan** or recently made any changes – specially in sep month?

Check what competitor is doing – bcoz one user has moved to competitor

Focus on customers with ‘High’ & ‘Med’ churn risk, check their LTV (to make priority list), complaint request, reach out to them through email, sms, calls and fix their issue.

Thank You